

THE PLATFORM

One control core. Power conversion at every scale.

Developed in the Smart Power Lab at UC Santa Cruz, OpenVVF separates a reusable dual-MCU control and safety core from the power hardware. Its PWM, protection, and sensing interfaces connect to smaller user-supplied stages, the implemented Size 2 inverter, the released Size 2 DC/DC module, and future higher-power chassis. The result is an open foundation for power-conversion systems without a new controller for every application.

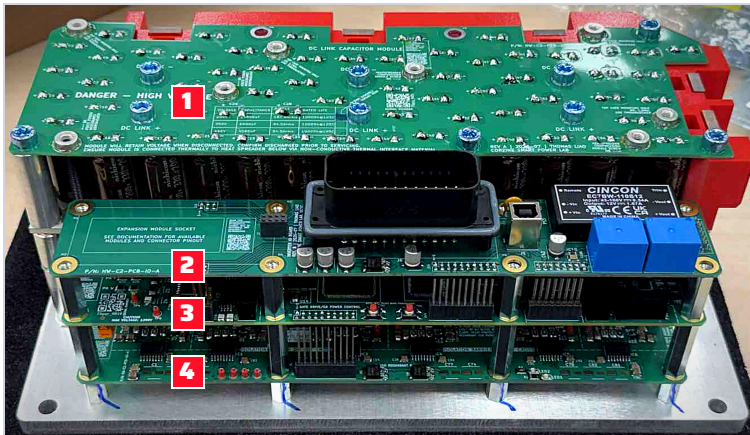


Fig. 1: The Size 2 stack, top to bottom: capacitor board, I/O board, control board, and gate-drive board.

1 DC-link capacitor board

Electrolytic cans are thermally bonded to the baseplate, with film and ceramic capacitors beneath. Multilayer ceramic capacitors (MLCCs) mount directly between the IGBT terminals. The short commutation loop limits switching overshoot and supports a future transition to SiC.

2 I/O board

USB-B, dual CAN FD, and an expansion socket for application I/O.

3 Control board, dual-MCU

FOC motor control runs on the main MCU. An independent safety coprocessor monitors every safety-critical signal and can bring the system to a safe state.

4 Gate-drive board

Isolated gate drive and sensing for the three IGBT half-bridges.

COOLING

Three IGBT half-bridge modules are conduction-cooled through an aluminum baseplate. Size 2 requires no liquid-cooling loop.

SOFTWARE AND WORKFLOW

The same control core is intended to run across power stages. It supports motor self-commissioning for PMSM and induction machines, the open-source RTE Studio model-based workflow, and the browser-based Telemetry Viewer. RTE Studio requires no MATLAB; a software-in-the-loop pipeline is in progress. The controls program has benchmarked ADRC and MPCC against the production FOC controller.



Fig. 2: Ceramic capacitors between the IGBT terminals. The commutation loop measures only a few millimeters.

MODULAR POWER BLOCKS

Scale down or up around one control core

CONTROL MODULE Dual-MCU · FOC motor control + independent safety coprocessor · USB-B, dual CAN FD, and expansion I/O · RTE Studio + Telemetry Viewer



Lower-power stages

Smaller or custom gate-drive stages with interface scaling

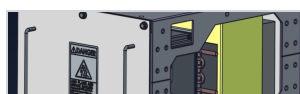
■ SCALABLE INTERFACE



Size 2 inverter

200–450 V class · 465 A continuous design rating

■ IMPLEMENTED, UNDER TEST



Size 2 DC/DC

15 kW · 450 V class · reuses Size 2 boards and power stage

■ RELEASED DESIGN



Size 3

Up to 1,200 V · 1,400 A · modular liquid-cooled enclosure

■ IN DEVELOPMENT

EVIDENCE

Bench-tested, on the record

200-450 V Size 2 DC-bus class by capacitor selection	465 A Continuous design rating · 600 A peak (60 s)	10 Published test reports, raw telemetry linked	PMSM + induction Instrument-verified motor self-commissioning
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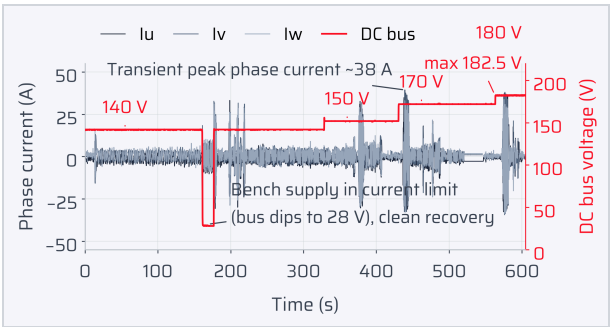


Fig. 3: 180 V power-stage bring-up. A Size 2 chassis (200 V-class DC link) drove a TECO MAX-IE3 10 HP induction motor as the bus stepped through 140, 150, 170, and 180 V. At approximately 160 s, the supply reached its current limit; after the limit was raised, the controller recovered without a reset or fault. Peak phase current was approximately 38 A; maximum bus voltage was 182.5 V.

TEST CAMPAIGN	SETUP	KEY RESULT
180 V power-stage bring-up	TECO 10 HP induction motor; bus stepped from 140 to 180 V	200 V-class operation confirmed; ≈38 A peak; clean fault recovery
One-hour no-load endurance	Size 2 + induction motor; 120 V DC; ≈60 Hz	60 minutes of continuous operation; zero faults; thermal baseline established
20-minute reversal stress at 180 V	±60 Hz every ≈45 s; passive cooling	26 reversals; zero faults; ≈41 A maximum
Self-commissioning accuracy	Inverter estimates versus LCR meter; PMSM + induction	Resistance error from -6.5 to +7.3 %; inductance error of +2.4 %

Six additional instrumented calibration reports support these results. All ten reports include raw telemetry and are revised publicly as validation continues.

SAFETY ENGINEERING

Six independent paths to the safe state

The control module is dual-MCU by design. The main MCU (**STM32H723ZG**) runs FOC motor control, plausibility checks, and fault handling. The safety coprocessor (**STM32G474RCTx**) independently monitors every safety-critical signal, cross-checks both CAN buses, and runs a challenge-response watchdog. Either processor can force the safe state.

Hardware PWM break (TIM1)	<100 ns	Shared driver RESET	<1 μs
1oo2 (one-out-of-two) gate-drive power kill	-10 μs	Independent coprocessor kill	<10 μs
Watchdog-forced main-MCU reset	-100 ms	Passive 3.3 V rail-loss pull-down	passive, no software

ASIL D design target for the highest safety goals, using ASIL B(D) decomposition: two independent ASIL B channels satisfy an ASIL D goal. ISO 26262 methodology is applied as a design input, not presented as a certification claim. The functional-safety roadmap includes fault-injection, thermal, and vibration campaigns.

17 hazards identified	>	14 safety goals	>	22 functional safety requirements	>	92 fault-injection tests planned	>	ISO 26262 + IEC 61800-5-2 safe-function mapping
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Safety engineered from the silicon to the sheet metal.

Support open power-conversion research

OpenVVF is developed in the Smart Power Lab, part of Electrical and Computer Engineering at UC Santa Cruz. Professor Keith Corzine leads the project, and Professor Leila Parsa is also a Smart Power Lab faculty contact. The project publishes hardware, firmware, safety engineering, and test evidence that others can study, build, and extend.

Because this work is university-based, funding and in-kind contributions are coordinated through UC Santa Cruz. Organizations can begin with Professor Corzine or Professor Parsa on technical fit; the university then determines the appropriate gift, sponsored research, or in-kind pathway.

CURRENT PROJECT SUPPORTERS



WAYS TO SUPPORT THE LAB

FLEXIBLE SUPPORT

Research gift

University-administered support for student research, components, fabrication, instrumentation, and early validation.

DEFINED SCOPE

Sponsored research

A formal project with an agreed technical scope, budget, schedule, and deliverables, reviewed and accepted by UC Santa Cruz.

EQUIPMENT AND SERVICES

In-kind contribution

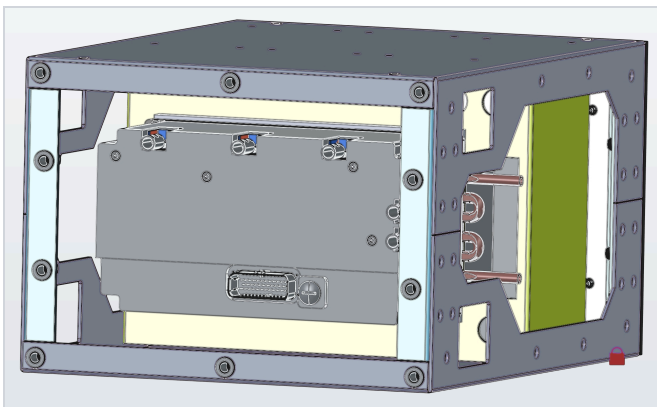
Power modules, components, fabrication, instrumentation, equipment loans, dynamometer time, or other test access.

SHARED TECHNICAL INTEREST

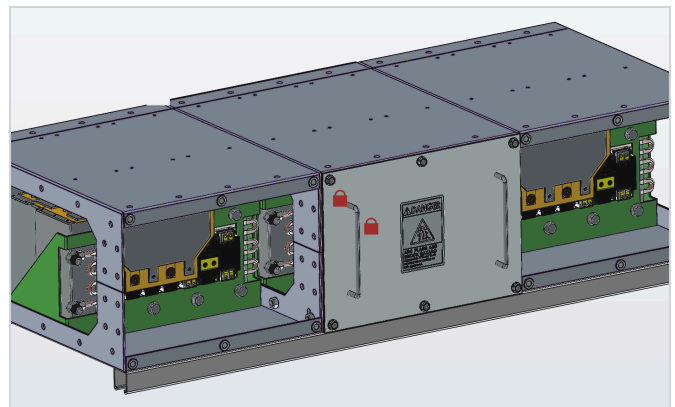
Research collaboration

Joint work on modular power stages, DC/DC conversion, controls, or validation, with the appropriate UCSC agreement.

How it starts: discuss technical alignment with Professor Corzine or Professor Parsa. UC Santa Cruz then identifies the correct administrative route and handles proposal, gift, equipment, or agreement requirements.



Size 2 DC/DC module: Released design rendering. The 15 kW converter reuses the Size 2 PCBs and power stage; orderable BOMs are available.



Size 3 high-power enclosure: Design rendering. The up-to-1,200 V, 1,400 A concept uses modular liquid-cooled bays; arc-flash containment analysis is in progress.

Start with a technical conversation. Support can advance Gen 7.1 hardware, loaded testing, vehicle integration, and reusable power-conversion modules.

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